

AI Gym & Fitness Assistant

Project Report — Streamlit-based Academic Prototype / MVP

1. Project Overview

AI Gym & Fitness Assistant is a modular fitness application designed to bring workout assistance, diet guidance, habit tracking, performance analysis, gym recommendations, smart gym settings, and conversational motivation into one dashboard. The implemented prototype focuses on demonstrating the core functionality of these modules through a Python and Streamlit application.

2. Objectives

- Provide basic computer-vision-based workout assistance and squat repetition counting.
- Generate general BMI, calorie, and meal suggestions according to the selected fitness goal.
- Track workout consistency and provide habit feedback.
- Provide fitness motivation through a Virtual Gym Buddy.
- Calculate a basic performance score from form, repetition quality, and consistency.
- Recommend gym/program types and generate a weekly fitness plan.
- Provide smart workout intensity, resistance, rest-time, and simulated equipment-status information.

3. Implemented Modules

Module	Implemented Functionality
AI Gym Trainer	MediaPipe pose detection, squat counting, knee-angle calculation and basic form feedback.
AI Diet Coach	BMI calculation, general calorie suggestion and goal-based meal suggestions.
Fitness Habit Tracker	Workout/missed-day tracking, consistency percentage, status and next-workout prediction.
Virtual Gym Buddy	Rule-based fitness conversation, motivation and workout guidance.
Performance Analyzer	Weighted performance score, performance level and feedback.
Gym Recommender & Planner	Goal/experience-based recommendations and weekly workout plan.
Smart Gym Assistant	Goal/experience-based intensity, resistance, rest time and simulated equipment status.

4. Technology Stack

Programming: Python **Interface:** Streamlit **Computer Vision:** OpenCV, MediaPipe **Data/Utilities:** Pandas, NumPy **Visualization:** Plotly **ML Library:** Scikit-learn included in the project environment.

5. Testing and Results

All seven implemented modules were tested through the Streamlit dashboard. The AI Gym Trainer was also tested with the webcam and successfully performed squat detection/repetition counting. The AI Diet Coach was tested for different dashboard goals, including Weight Loss and Muscle Gain. The other modules produced their expected dashboard outputs without reported runtime errors.

6. Project Limitations

This submission is an academic prototype/MVP rather than a production fitness platform. The Smart Gym Assistant uses simulated equipment availability rather than live IoT devices. The Virtual Gym Buddy is rule-based rather than an LLM chatbot. The application does not currently implement a persistent database, cloud storage, or a React/FastAPI production backend. Diet and calorie outputs are general guidance and are not medical advice.

7. Conclusion

The implemented AI Gym & Fitness Assistant demonstrates how multiple fitness-support features can be integrated into a single Streamlit dashboard. The prototype successfully combines computer vision, rule-based recommendations, habit analysis, performance scoring, planning, and conversational support. The modular structure also provides a foundation for future extensions such as real IoT integration, persistent user data, advanced ML models, LLM-based

conversation, and production deployment.

Repository: AI-Gym-Fitness-Assistant (GitHub) — <https://github.com/adityakumarlenka15-byte/AI-Gym-Fitness-Assistant>